DO NOW — RECALL

Lesson 095 · CPU Pipelining: Overlap and Hazards

Name: _____ Class: _____ Date: _____

1. What is pipeline throughput?

2. What is a hazard?

3. Which CPU feature may reduce RAM accesses?

4. A D flip-flop has D=0 and Q=1 just before the active rising edge. What is Q after?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Rate of completed instructions.

2. Model answer

Dependency/resource/control issue that interrupts pipeline.

3. Model answer

Cache.

4. Model answer

Q becomes0, sampled at the active edge.

By the end, you can…

• Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

• With four independent instructions and three one-cycle stages, ideal completion is 3+4−1=6 cycles; serial non-overlap takes 4×3=12.

• Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

Vocabulary — know these terms

pipeline — Stages overlapping different instructions.

throughput — Completed work per time.

data hazard — Dependency on unavailable result.

control hazard — Branch uncertainty interrupting fetch.

NEW LEARNING

Read, annotate and explain

Observe

Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

Explain

With four independent instructions and three one-cycle stages, ideal completion is $3+4-1=6$ cycles; serial non-overlap takes $4\times3=12$.

Evaluate

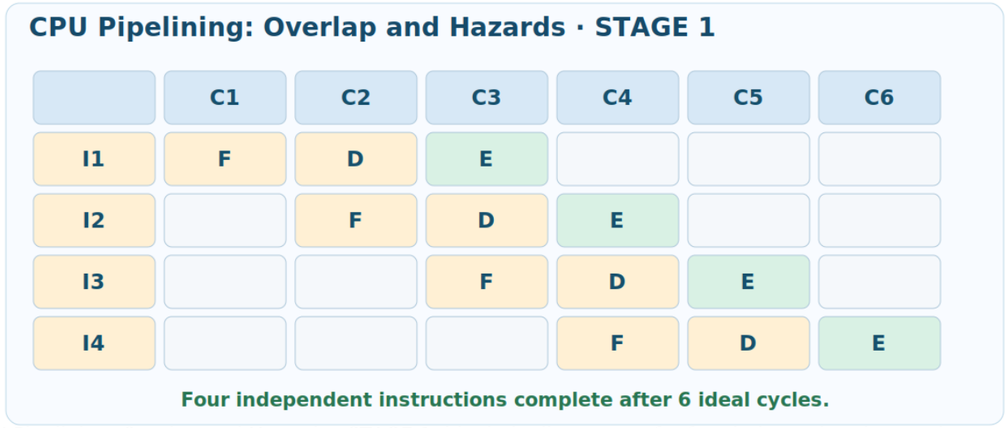
Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

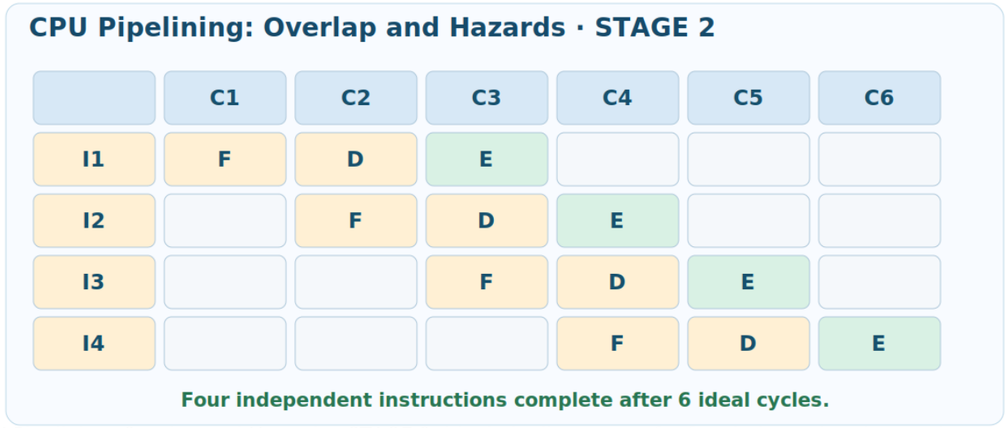
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU Pipelining: Overlap and Hazards · STAGE 1; ideal pipeline stages for four independent instructions

New Learning



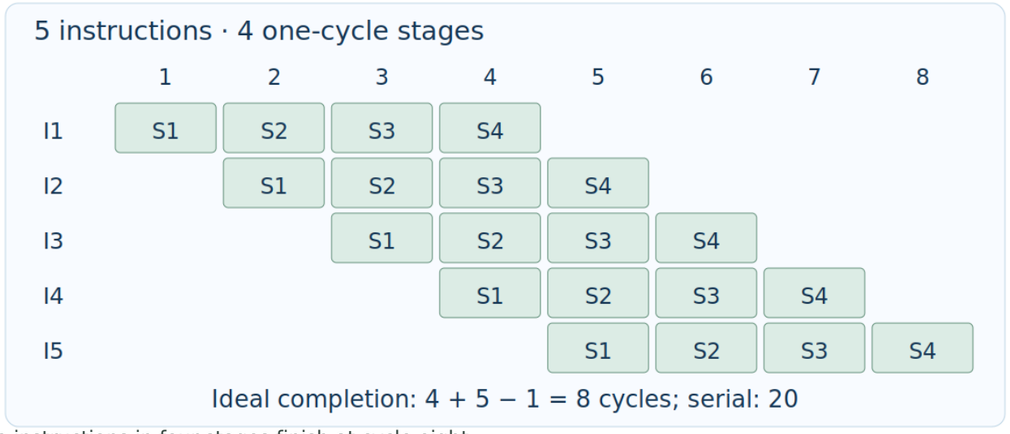
CPU Pipelining: Overlap and Hazards · STAGE 2; ideal pipeline stages for four independent instructions

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Five independent instructions use a four-stage ideal pipeline. Calculate completion.

4+5−1=8 cycles, compared with 4×5=20 serial stage cycles.



Five instructions in four stages finish at cycle eight

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

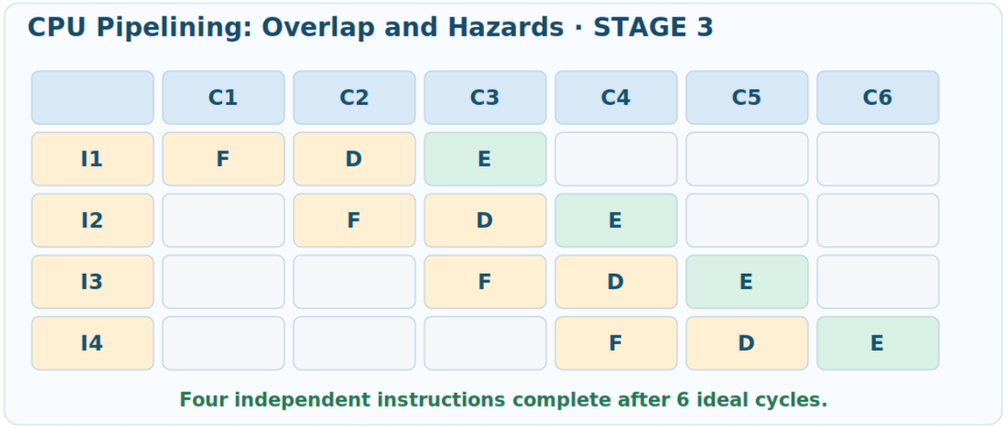
Guided 1 — CHECK

Explain the mechanism
4+5−1=8 cycles, compared with 4×5=20 serial stage cycles.
Assumes equal one-cycle stages, no resource conflicts or stalls.

New Learning · original lesson diagrams

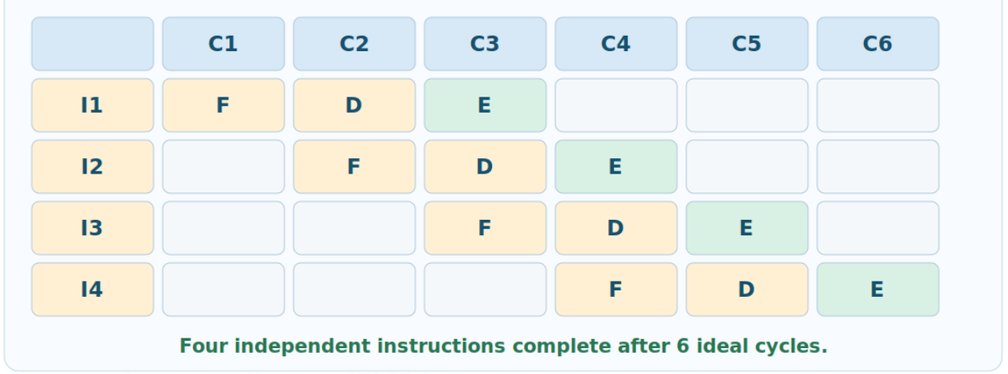
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU Pipelining: Overlap and Hazards · STAGE 3; ideal pipeline stages for four independent instructions

New Learning



CPU Pipelining: Overlap and Hazards · STAGE 4; ideal pipeline stages for four independent instructions

I DO — WORKED EXAMPLE

Follow the worked method

Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.

Explain the mechanism

2+3−1=4 cycles; serial baseline 2×3=6 cycles.

The second task starts stage one while first advances to stage two.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A pipeline makes each stage finish in less than one cycle by default.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

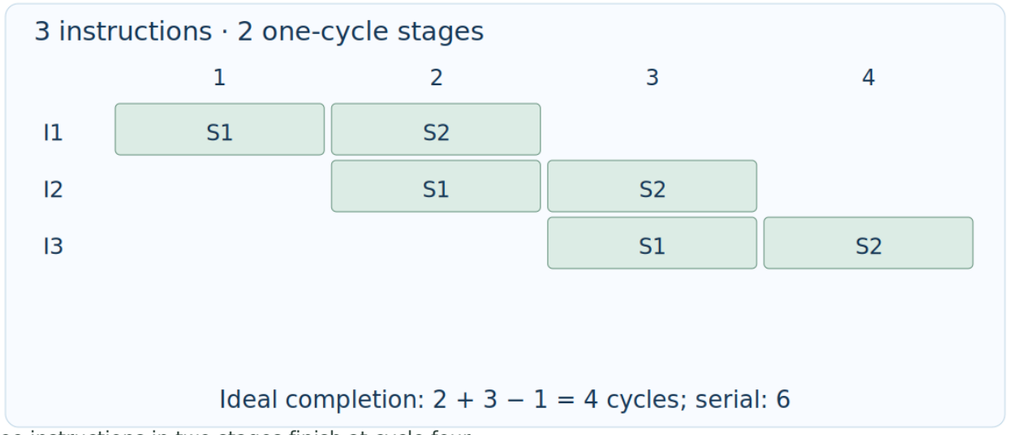
Explain why branch and data hazards can reduce actual improvement, distinguishing latency and throughput. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.

2+3−1=4 cycles; serial baseline 2×3=6 cycles.



Three instructions in two stages finish at cycle four

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Five independent instructions use a four-stage ideal pipeline. Calculate completion.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Show cycles for I1-I3 in a 3-stage F-D-E pipeline, and finish cycle of I3. [6]

Question 2

Compute ideal pipeline versus serial completion for 6 instructions and 3 stages. [6]